

A hypothetical pulse is travelling along positive x direction on a taut string. The speed of the pulse is 10 cm s^{-1} . The shape of the pulse at $t = 0$ is given as

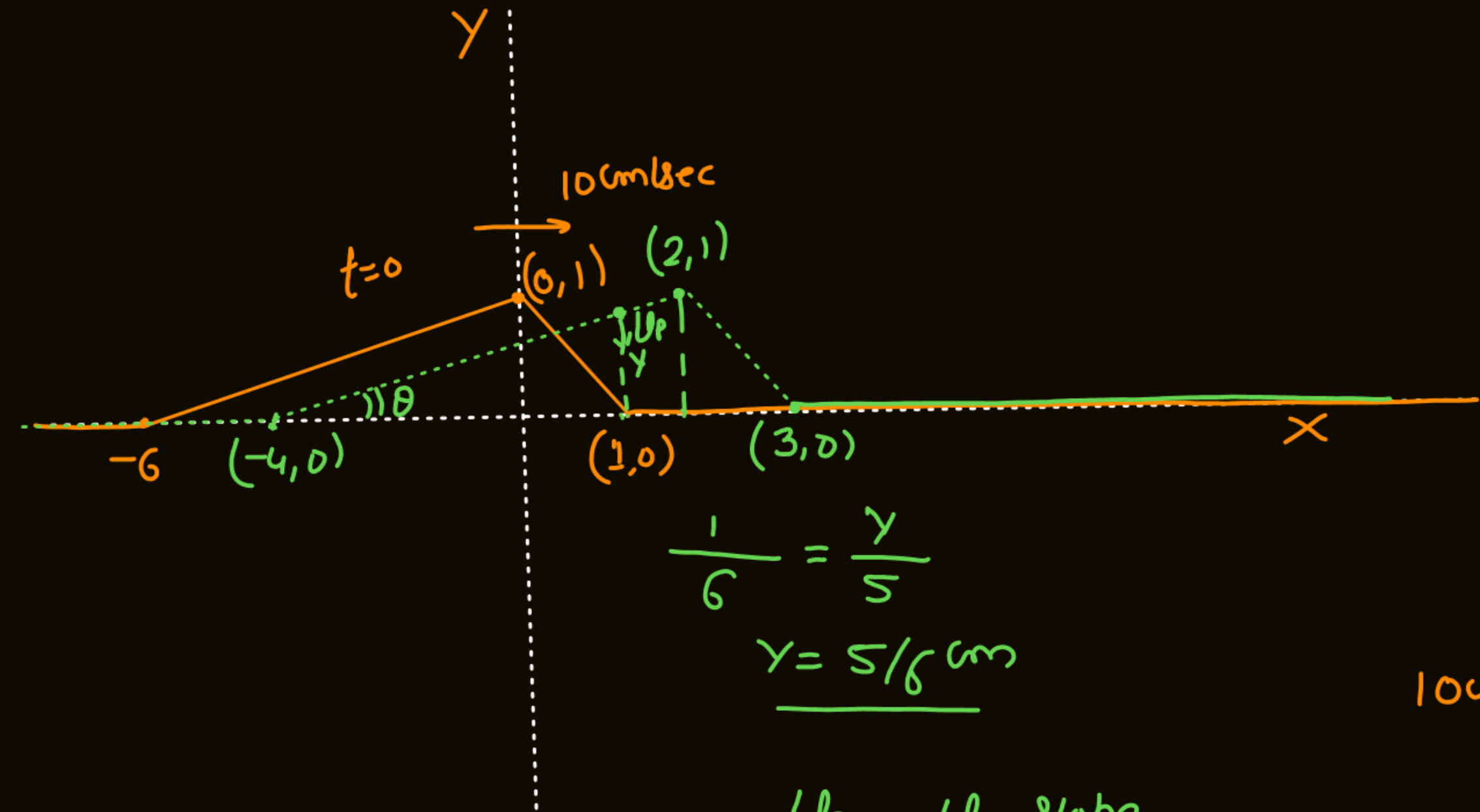
$$y = \frac{x}{6} + 1 \quad \text{for } -6 < x \leq 0$$

$$= -x + 1 \quad \text{for } 0 \leq x < 1$$

$$= 0 \quad \text{for all other values of } x$$

x and y are in cm .

- Find the vertical displacement of the particle at $x = 1 \text{ cm}$ at $t = 0.2 \text{ s}$
- Find the transverse velocity of the particle at $x = 1 \text{ cm}$ at $t = 0.2 \text{ s}$.



$$\frac{1}{6} = \frac{y}{5}$$

$$y = \frac{5}{6} \text{ cm}$$

$$U_p = -U_w \text{ slope}$$

$$= -(10 \text{ cm/sec}) \times \frac{1}{6} = -\frac{5}{3} \text{ cm/sec}$$

$$10 \text{ cm/sec} \times 0.2 \text{ sec}$$

$$\underline{2 \text{ cm}}$$

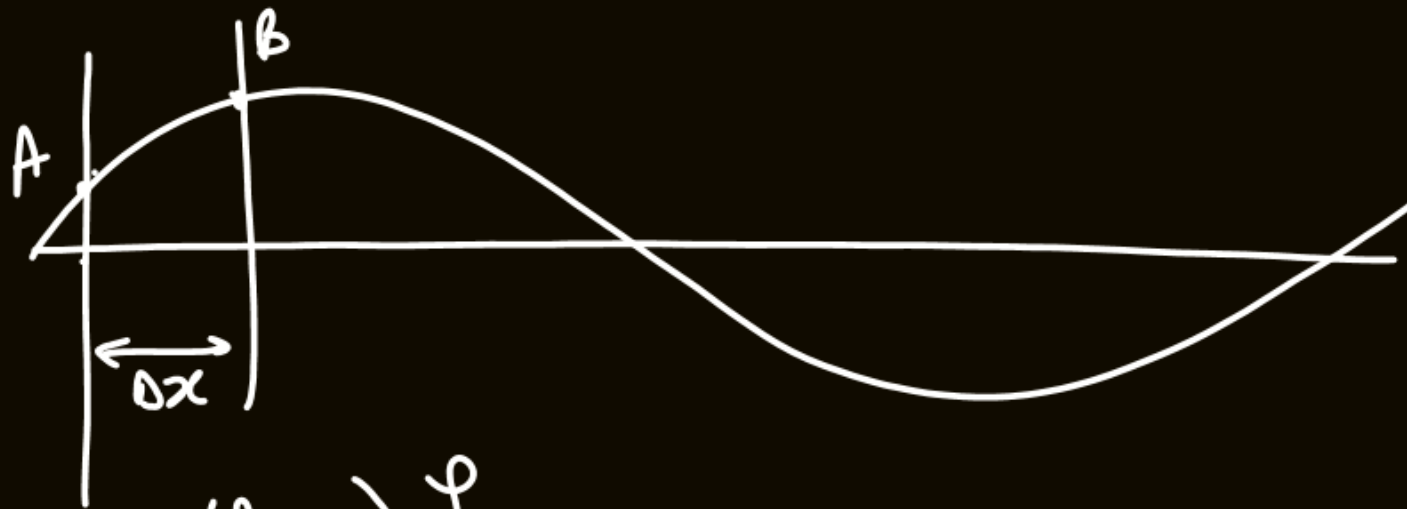
$$v = \sqrt{\frac{57.6}{0.1}} = \underline{24 \text{ m/s}}$$

$$\lambda \rightarrow 25$$

$$\Delta x \rightarrow \frac{25}{\lambda} \Delta x = \Delta \phi$$

$$\Delta \phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{\lambda} \times 5 \text{ cm}$$

$$\lambda = 60 \text{ cm} = 0.6 \text{ m}$$



$$v = \lambda f$$

$$f = \frac{v}{\lambda} = \frac{24}{0.6} = 40 \text{ Hz}$$

$$T = \frac{1}{f} = \frac{1}{40} \text{ sec}$$

A transverse harmonic wave travels along a taut string having a tension of 57.6 N and linear mass density of 100 g/m. Two points A and B on the string are 5 cm apart and oscillate with a phase

difference of $\frac{\pi}{6}$. How much does the phase of oscillation of point A change in a time interval of 5.0 ms?

$$T \rightarrow 25$$

$$\Delta t \rightarrow \frac{2\pi}{T} \Delta t = \Delta \phi$$

$$\Delta \phi = \frac{2\pi}{40} \times 5 \times 10^{-3}$$

$$= 40 \times 2\pi \times 5 \times 10^{-3}$$

$$= \underline{\underline{\frac{2\pi}{5} \text{ rad}}}$$

$$A = 2 \text{ mm}$$

$$f = 50 \text{ Hz}$$

$$\lambda = 4 \text{ m}$$

$$v_w = \lambda f$$

$$= 200 \text{ m/s}$$

$$v_w = \sqrt{\frac{T}{\mu}}$$

$$T = \mu v_w^2$$

$$T = 0.01 \times 200 \times 200$$

$$T = 400 \text{ N}$$

$$\mu = \frac{0.1 \times 10^{-3} \text{ kg}}{10^{-2} \text{ m}} = 0.01 \text{ kg/m}$$



$$E = P_{\text{avg}} t$$

$$E = (2\pi^2 f^2 A^2 \mu v) t$$

$$= 2\pi^2 \times (50)^2 \times (2 \times 10^{-3})^2$$

$$\times 0.01 \times 200 \times 1$$

$$= 2\pi^2 \times \frac{2500}{10000} \times 4 \times 10^{-6} \times 2 \times 10^{-2}$$

$$= \frac{4\pi^2}{100} = \frac{\pi^2}{25} \text{ Joules}$$

A sinusoidal wave travels along a taut string of linear mass density 0.1 g/cm . The particles oscillate along y -direction and the disturbance moves in the positive x -direction. The amplitude and frequency of oscillation are 2 mm and 50 Hz respectively. The minimum distance between two particles oscillating in the same phase is 4 m .

(a) Find the tension in the string.

(b) Find the amount of energy transferred through any point of the string in one second.

(c) If it is observed that the particle at $x = 2 \text{ m}$ is at $y = 1 \text{ mm}$ at $t = 2 \text{ s}$, and its velocity is in positive y -direction, then write the equation of this travelling wave.

$$y = A \sin(\omega t - kx + \theta)$$

$$y = 2 \text{ mm} \sin\left(100\pi t - \frac{\pi}{2}x + \theta\right)$$

$$1 = 2 \sin(200\pi - \pi + \theta)$$

$$\frac{1}{2} = \sin(199\pi + \theta)$$

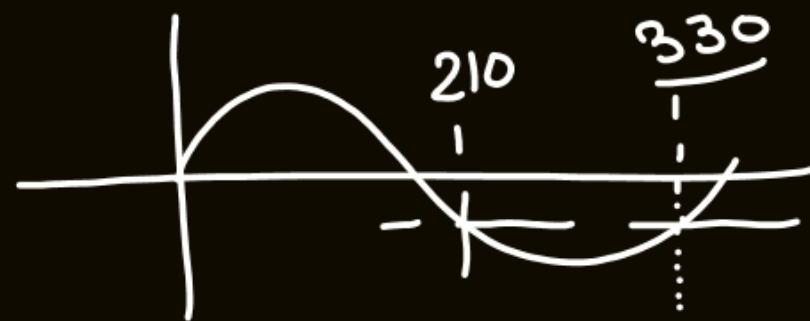
$$= \sin(\underline{198\pi} + \pi + \theta)$$

$$= \sin(\pi + \theta)$$

$$= -\sin\theta$$

$$\boxed{\sin\theta = -\frac{1}{2}}$$

$$y = 2 \text{ mm} \sin\left(100\pi t - \frac{\pi}{2}x + 210^\circ\right)$$



$$y = 2 \text{ mm} \sin\left(100\pi t - \frac{\pi}{2}x + \theta\right)$$

$$v_p = \frac{\partial y}{\partial t} = 2 \times 100\pi \cos\left(100\pi t - \frac{\pi}{2}x + \theta\right) \quad \underline{\underline{\theta = 210^\circ}}$$

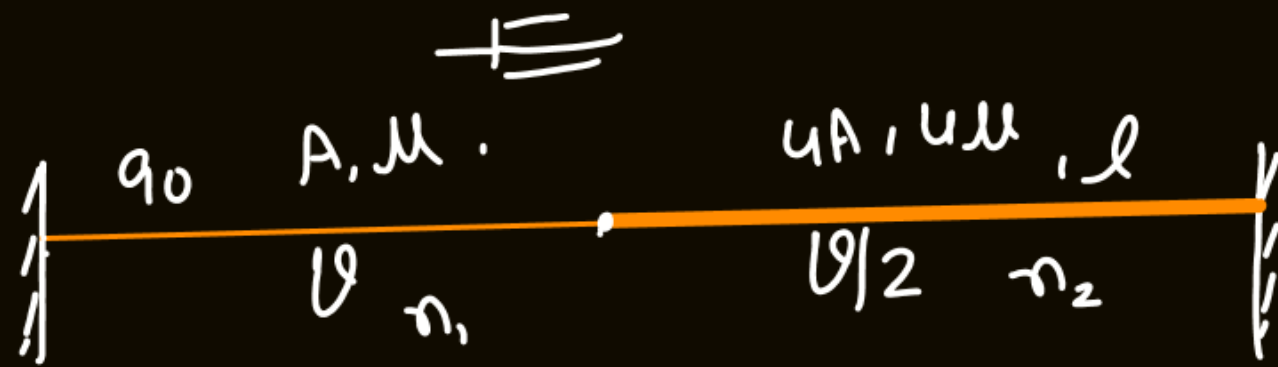
$$v_p = + = () \cos(200\pi - \pi + \theta)$$

$$+ = () \cos(\pi + \theta)$$

$$+ = - () \cos\theta$$

$$\cos\theta = - \quad \text{cos } 210^\circ = - \quad \text{cos } 330^\circ = +$$

Two strings of same material are joined to form a large string and is stretched between rigid supports. The diameter of the second string is twice that of the first. It was observed in an experiment that the whole string was oscillating in 4 loops with a node at the joint. Find the possible lengths of the second string if the length of first string is 90 cm.



$$\mu = \frac{A}{l}$$

$$v = \sqrt{\frac{T}{\mu}}$$

$$f = \frac{n_1 v}{2 \times 90} = \frac{n_2 v/2}{2 \times l}$$

$$\frac{n_1}{45} = \frac{n_2}{l}$$

$$\frac{1}{45} = \frac{3}{l} \Rightarrow l = 135 \text{ cm}$$

$$\frac{2}{45} = \frac{2}{l} \Rightarrow l = 45 \text{ cm}$$

$$\frac{3}{45} = \frac{1}{l} \Rightarrow l = 15 \text{ cm}$$

$$n_1 + n_2 = 4$$

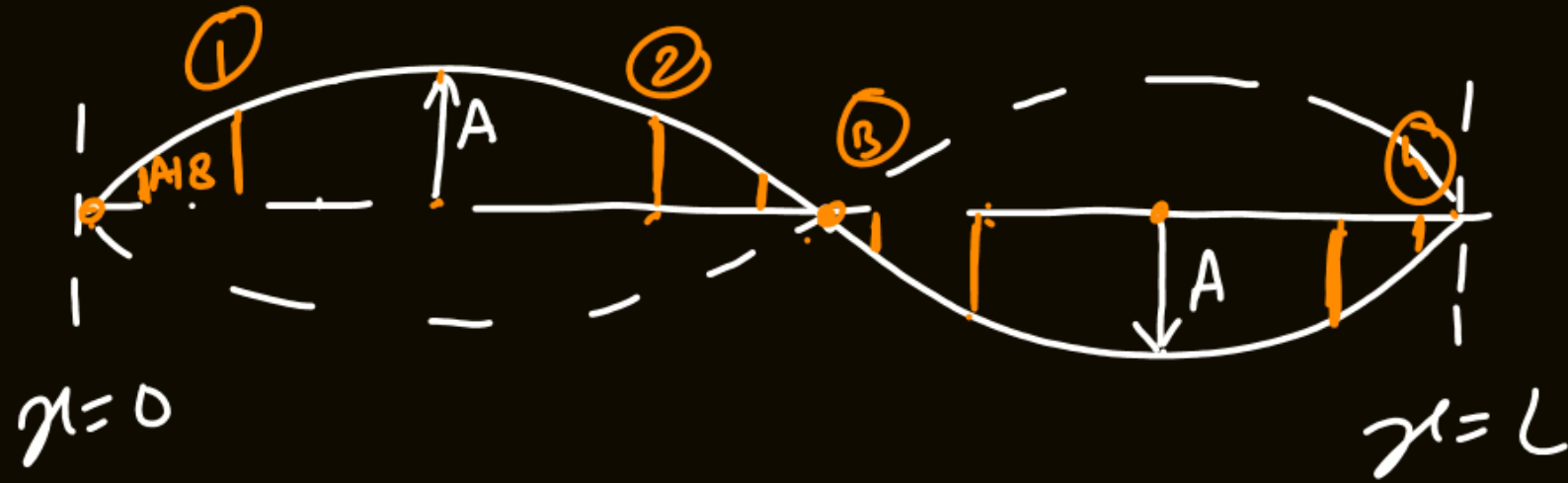
① $n_1 = 1 \quad n_2 = 3$

② $n_1 = 2 \quad n_2 = 2$

③ $n_1 = 3 \quad n_2 = 1$

A string, of length L , clamped at both ends is vibrating in its first overtone mode. Answer the following questions for the moment the string looks flat

- Find the distance between two nearest particles each of which have half the speed of the particle having maximum speed.
- How many particles in the string have one eighth the speed of the particle travelling at highest speed?



$$v_p = A\omega$$

$$x = \frac{L}{12}, \frac{5L}{12}, \frac{7L}{12}, \frac{11L}{12} \quad \sin \frac{2\pi}{L} x = \pm \frac{1}{2}$$

$$\frac{4L}{12}, \frac{2L}{12}, \frac{L}{6}$$

$$\text{or } \frac{2\pi}{L} x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$A' = A \sin\left(\frac{2\pi}{L} x\right)$$

$$A' = A \sin\left(\frac{2\pi}{L} x\right)$$

$$\pm \frac{A}{2} = A \sin\left(\frac{2\pi}{L} x\right)$$

$$f_0 = \frac{v}{2l} = \frac{\sqrt{\frac{T}{\mu}}}{2l}$$

$$\eta_1 f_0 = \frac{\sqrt{\frac{1.96T}{\mu}}}{2 \times 0.65l}$$

$$\frac{\eta_1}{\eta_2} = \frac{\frac{1.4 \times 1.3l}{0.65 \times 0.8}}{2}$$

$$= \underline{3.5}$$

$$\eta_2 f_0 = \frac{\sqrt{\frac{0.64T}{\mu}}}{2 \times 1.3l}$$

Fundamental frequency of a stretched sonometer wire is f_0 . When its tension is increased by 96% and length decreased by 35%, its fundamental frequency becomes $\eta_1 f_0$. When its tension is decreased by 36% and its length is increased by 30%, its fundamental frequency becomes $\eta_2 f_0$.

Find $\frac{\eta_1}{\eta_2}$.



$$T = Mg$$



$$f = \frac{n \sqrt{\frac{Mg}{\mu}}}{2 \times 2}$$

$$\checkmark f = \frac{n \sqrt{\frac{25 \times 10}{10^{-3}}}}{2 \times 2}$$

$$\checkmark f = \frac{(n+1) \sqrt{\frac{16 \times 10}{10^{-3}}}}{2 \times 2}$$

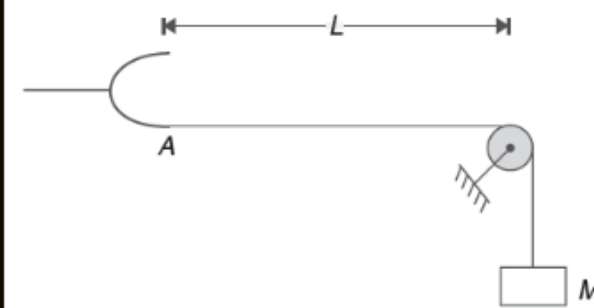
$$n \times 5 = (n+1) \times 4$$

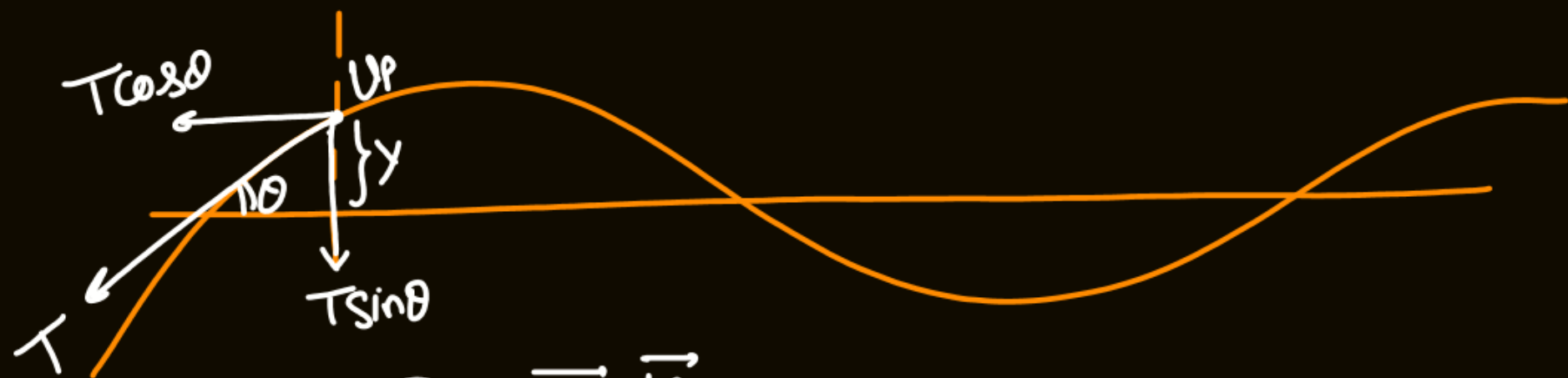
$$n = 4$$

$$f = \frac{4 \sqrt{25 \times 10^4}}{4}$$

$$f = 500 \text{ Hz}$$

The linear mass density of the string shown in the figure is $\mu = 1 \text{ g/m}$. One end (A) of the string is tied to a prong of a tuning fork and the other end carries a block of mass M . The length of the string between the tuning fork and the pulley is $L = 2.0 \text{ m}$. When the tuning fork vibrates, the string resonates with it when mass M is either 16 kg or 25 kg. However, standing waves are not observed for any other value of M lying between 16 kg and 25 kg. Assume that end A of the string is practically at rest and calculate the frequency of the fork.





$$P = \vec{T} \cdot \vec{v}$$

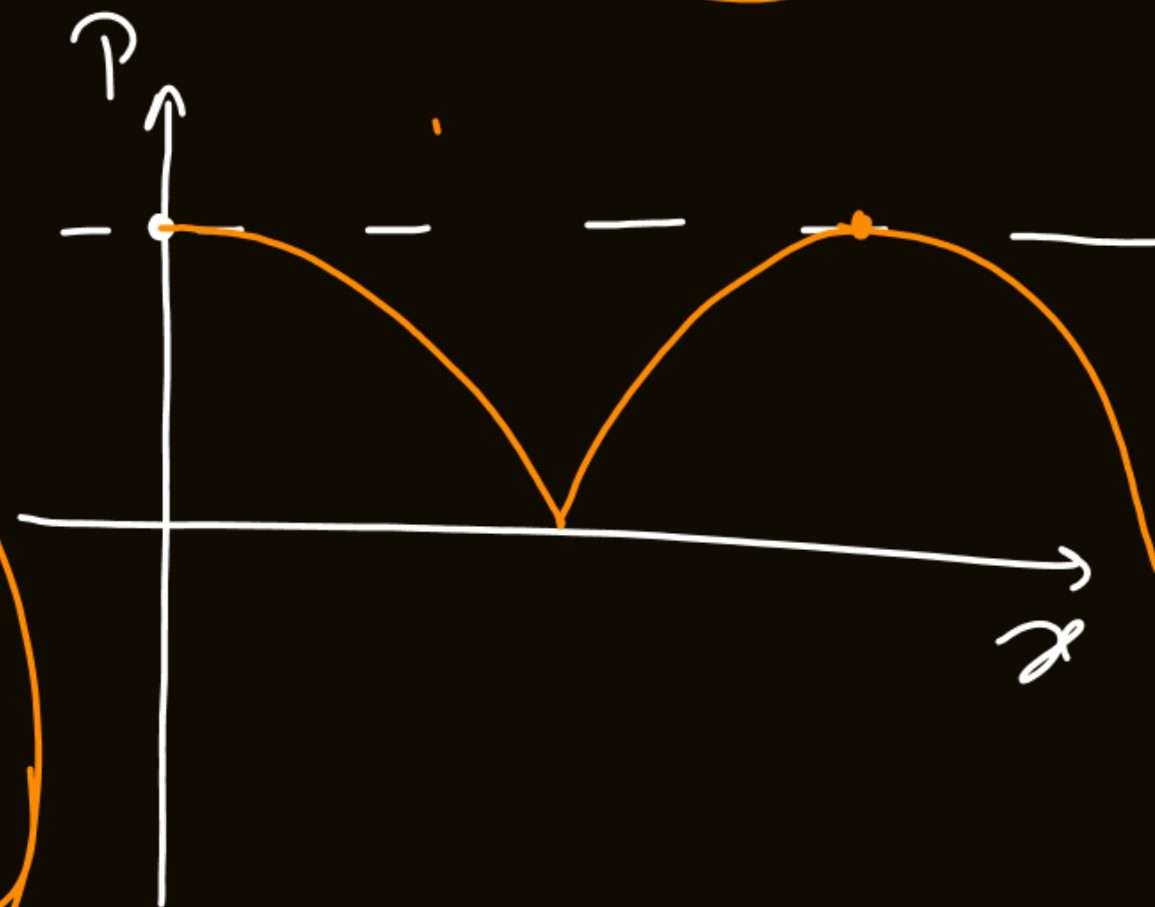
$$= (-T \cos \theta \hat{i} - T \sin \theta \hat{j}) \cdot \frac{\partial y}{\partial t} \hat{j}$$

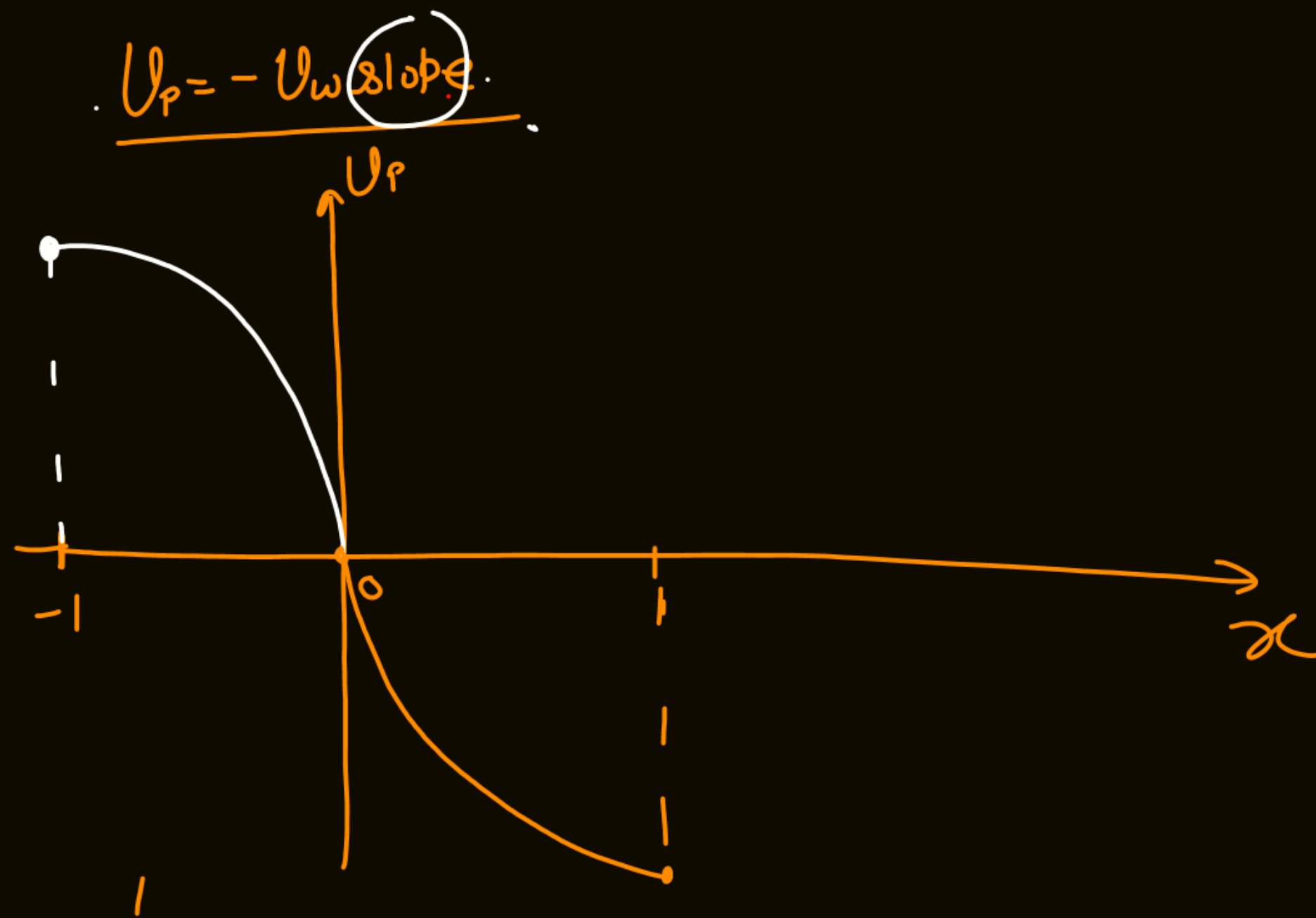
$$= -T \sin \theta \cdot \frac{\partial y}{\partial t}$$

$$= -T \tan \theta (v_p)$$

$$= -T \frac{\partial y}{\partial x} \left(-v_\omega \frac{\partial y}{\partial x} \right) = T v_\omega \left(\frac{\partial y}{\partial x} \right)^2$$

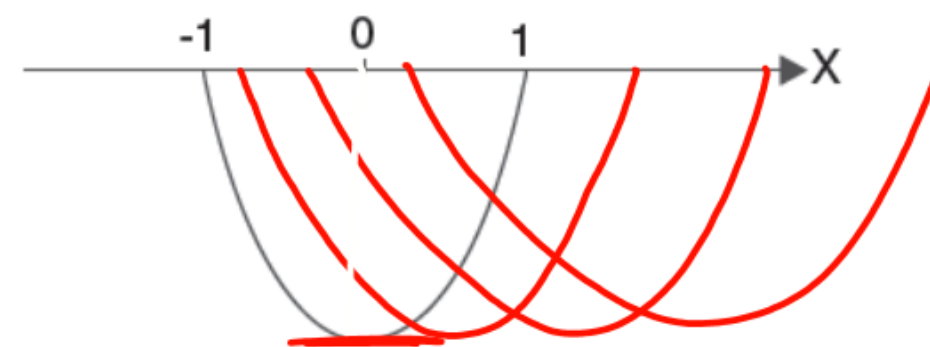
The figure shows y (transverse displacement) vs x (position) graph for a sinusoidal wave travelling along a stretched string. P is power transmitted through a cross section of the string at the instant shown. Plot the graph of P versus x .



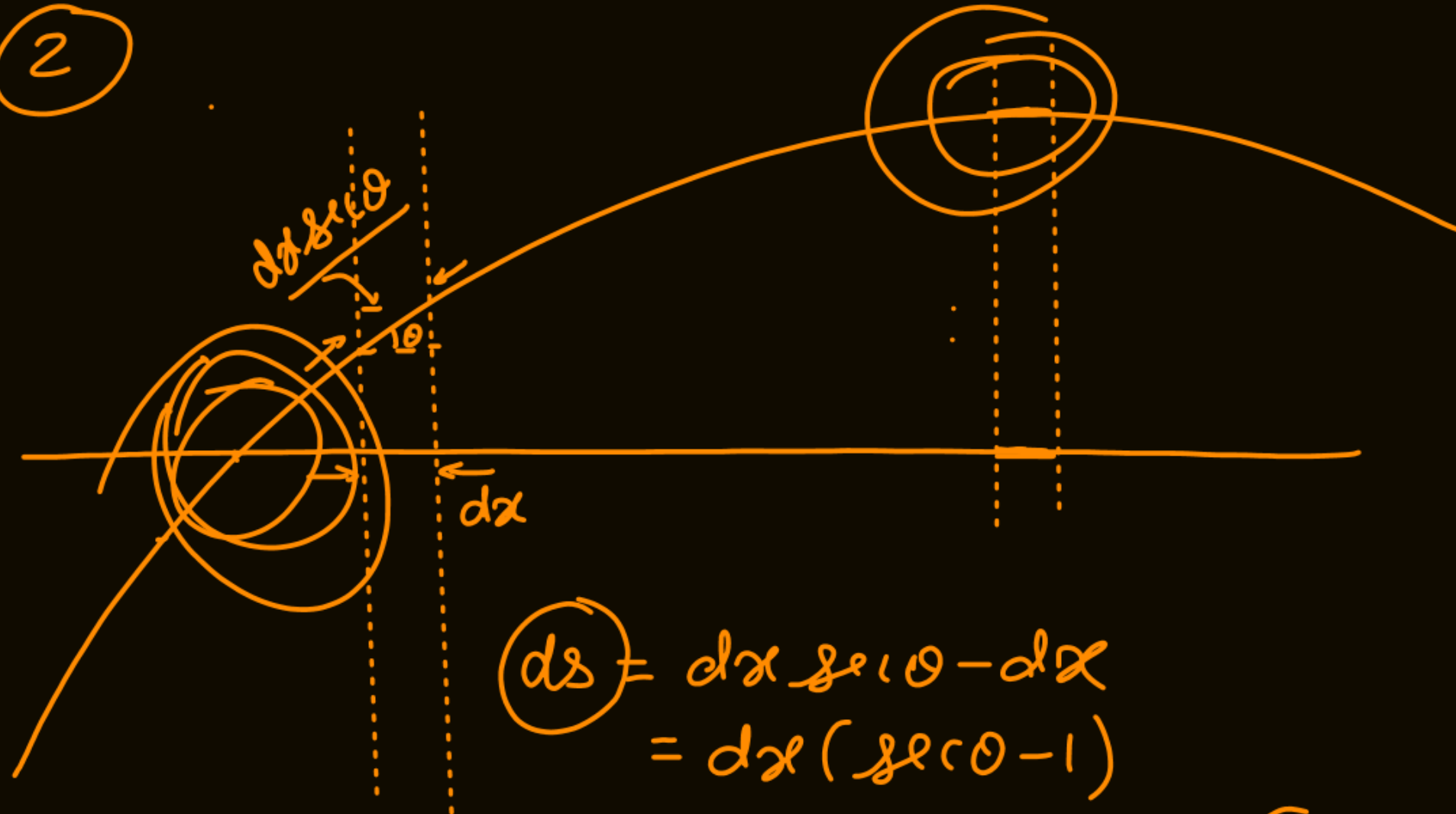


The figure shows the snapshot at time $t = 0$ of a transverse pulse travelling on a string in positive x direction.

- Sketch the pulse at a slightly later time.
- With the help of the given sketch draw a graph of velocity of each string segment versus position. Take upward direction as positive.



(i) (2)



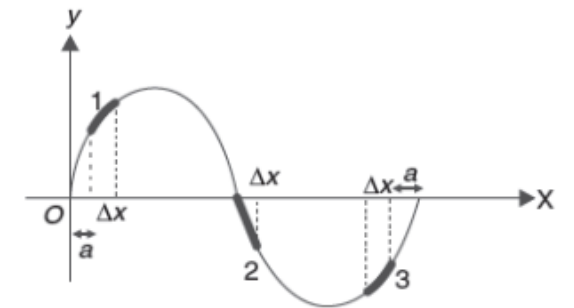
$$\begin{aligned} ds &= dx \sec \theta - dx \\ &= dx (\sec \theta - 1) \end{aligned}$$

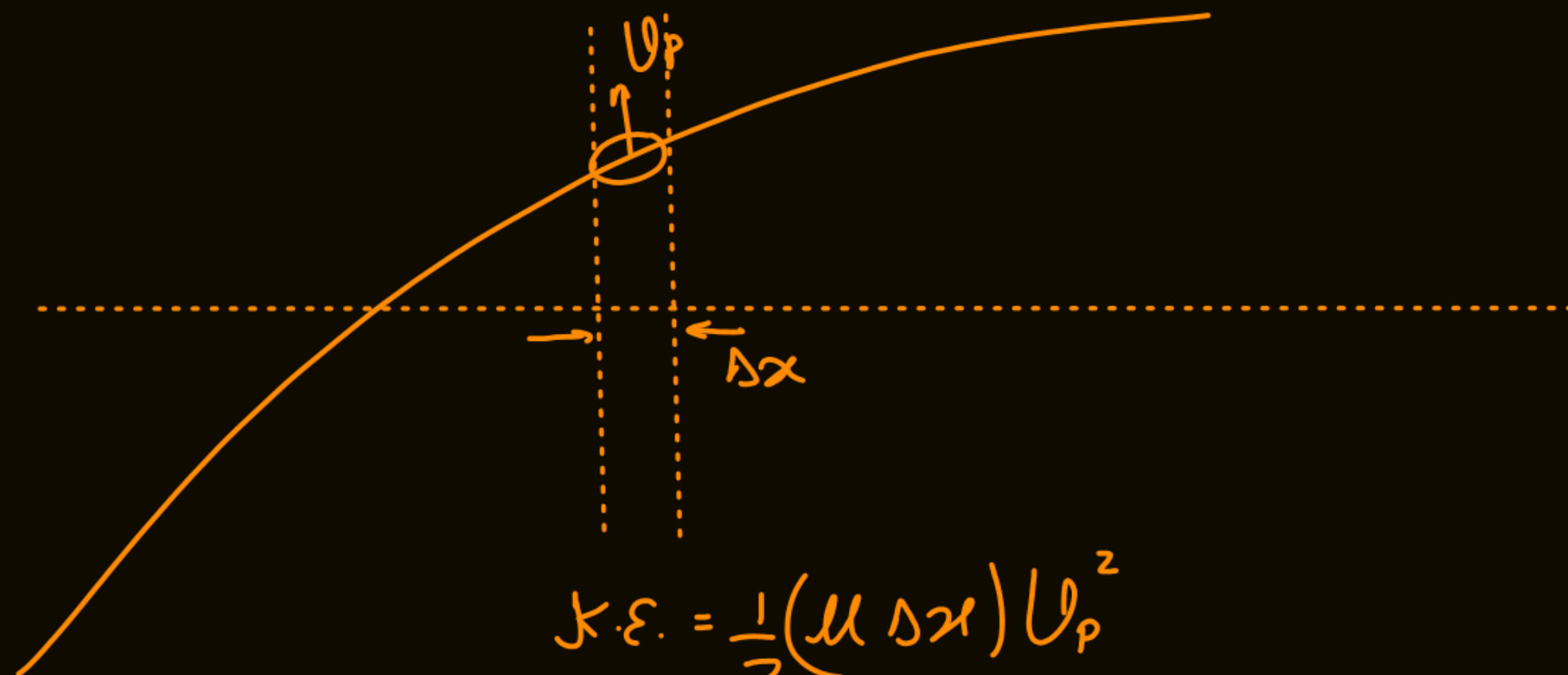
near 0 $\sec \theta \approx 1 + \frac{1}{2} \theta^2$ $\therefore ds \approx \frac{1}{2} dx \theta^2$ Energy

The figure shows the $y - x$ graph at an instant for a small amplitude transverse wave travelling on a stretched string. Three elements (1, 2 and 3) on the string have equal original lengths ($= \Delta x$). At the given instant-

- which element (among 1, 2 and 3) has largest kinetic energy?
- which element has largest energy (i.e., sum of its kinetic and elastic potential energy)

- Prove that energy per unit length $\frac{\Delta E}{\Delta x}$ of the string is constant everywhere equal to $T \left(\frac{\partial y}{\partial x} \right)^2$ where T is tension the string.





$$K.E. = \frac{1}{2} (\mu \Delta x) U_p^2$$

$$\frac{K.E.}{\Delta x} = \frac{1}{2} \mu \left(U_w \frac{\partial y}{\partial x} \right)^2 \Rightarrow \frac{K.E.}{\Delta x} = \frac{1}{2} \mu U_w^2 \left(\frac{\partial y}{\partial x} \right)^2$$

$$U_w = \sqrt{\frac{T}{\mu}}$$

$$T = \mu U_w^2$$

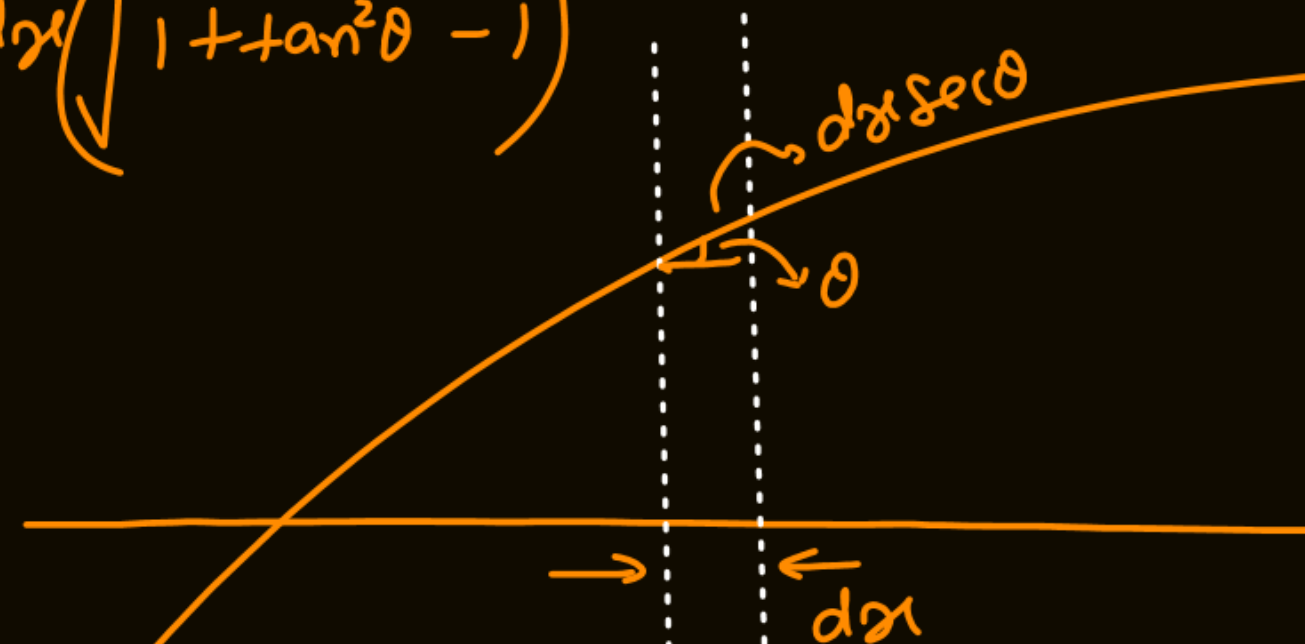
$$\frac{\Delta U}{\Delta x} = \frac{1}{2} T \left(\frac{\partial y}{\partial x} \right)^2$$

$$\frac{\Delta U}{\Delta x} = \frac{1}{2} T \left(\frac{\partial y}{\partial x} \right)^2$$

$$\frac{\Delta E}{\Delta x} = T \left(\frac{\partial y}{\partial x} \right)^2$$

$$dl = dx(\sec\theta - 1)$$

$$= dx(\sqrt{1 + \tan^2\theta} - 1)$$



$$dl = dx \left\{ (1 + \tan^2\theta)^{\frac{1}{2}} - 1 \right\}$$

$$= dx \left\{ 1 + \frac{1}{2} \tan^2\theta - 1 \right\}$$

$$dl = \frac{1}{2} (dx) \tan^2\theta$$

$$\Delta U = T dl$$

$$= \frac{T}{2} dx \tan^2\theta$$

$$\frac{\Delta U}{dx} = \frac{1}{2} T \left(\frac{\partial y}{\partial x} \right)^2$$